

The impact of sentiment analysis on moral judgment: Analyzing r/AITA subreddit comments

Project Report

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1 Introduction

This research aims at contributing to a deeper understanding of the relationship between moral verdicts and emotional content in online communities, such as the types of conflicts that are more likely to elicit strong negative or positive sentiments, specifically within the context of the `r/AITA` subreddit. By analysing the relation between the emotional intensity of comments and the outcomes of sentiment classification as well as the `YTA/NTA` labelling, we seek to uncover whether emotional content poses a challenge or enhances the effectiveness of sentiment analysis models.

The community `r/AITA` (Am I The Asshole) is a specific public online forum within the social news aggregation and discussion website Reddit. This so called *subreddit* is a virtual place where people share their real-life conflicts and ask for public judgements on whether they are in the wrong or not. Thus the community votes on the writer's behaviour by replying with short to elaborate comments and stating their subjective verdicts, which are mostly either `YTA` (*You're the asshole*) or `NTA` (*Not the asshole*). While counting over 7.5 million subscribers as of May 2023, the motivation for analyzing `r/AITA`'s comments stems from the growing popularity of this subreddit and the need to understand the sentiments expressed by its users. [Red13]

First, we establish a baseline classifier using Logistic Regression, which predicts the `YTA/NTA` label for a given document. This baseline classifier serves as a starting point for our research. To enhance the performance of our baseline classifier, we explore a range of advanced models such as *BERT* (Bidirectional Encoder Representations from Transformers). By employing these models, we aim to identify a model that effectively addresses our research question and captures the relationship between `YTA/NTA` labelling and emotional comments or sentiment analysis, respectively.

In tandem with this, Welch et al. addressed a similar research question [Wel+22]. As part of their work, they analyzed the influence of strong negative emotions on their model's accuracy and F1-Scores [Wel+22]. For that, the authors fine-tuned an *SBERT* model (Sentence Bert) on the binary prediction task [RG19]. Also, they used different aspects of conflict and developed a suchlike clustering on a subset of the *User's Perception* dataset [Wel+22]. In contrast, we take a look at both sides of the spectrum: Therefore we try to unravel the influence that strong positive as well as strong negative emotions have on our own trained classifiers. Also, we choose to explore another approach by using pretrained sentiment classifiers as a mean to filter our dataset such that mild emotional comments are preserved.

2 Methodology

This chapter presents the methodology employed for the development of a `YTA/NTA` classifier, incorporating a comprehensive analysis of the writer's sentiment as a potential influencing factor. To achieve this, we employ a series of steps that involve checking for a dependency between label and sentiment, building different comment classifiers, leveraging pretrained sentiment analysis models and comparing and analyzing both steps' results.

As a first step, we test whether a dependence between the stochastic variable label (`YTA,NTA`) and a sentiment label(`negative,neutral,positive`) exists. This is done to examine whether the sentiment alone fulfill the role of classifying a comment to a label. The pretrained models used for sentiment classification are *VADER* and *Twitter-roBERTa-base for Sentiment Analysis*. *VADER* (Valence Aware Dictionary and sEntiment Reasoner) is a widely adopted lexicon and rule-based sentiment analysis tool specifically designed for analyzing social media text [HG14]. *Twitter-roBERTa-base for Sentiment Analysis* is a pretrained sentiment analysis model that has been specifically designed for sentiment analysis in social media text [Car22].

Next we establish a baseline for classifying comments. For that purpose we employ a Logistic Regression, utilizing *TF-IDF* (Term Frequency-Inverse Document Frequency) as the embedding. We try to improve upon this baseline by using two different approaches: the first approach consists of replacing the model and embedding with more sophisticated techniques. For the second approach we incorporate the writers sentiment, by adjusting the data set used to train the respective model.

In line with our first approach, we combine the following models with GloVe as the embedding: a *MLP* (Multi-Layer Perceptron), a *GRU* (Gated Recurrent Unit) and a model using *LR* (Logistic Regression). Lastly, we employ *BERT*, a powerful transformer-based model, as the embedding in conjunction with an MLP as the classifier. BERT has demonstrated remarkable success in various natural language processing tasks, and we try to leverage its capabilities to improve the classification performance [Dev+18].

According to our second approach, we attempt to possibly further increase the performance of our classifiers by training them with a filtered dataset. The filtering process involves using the pretrained sentiment classifiers to identify and remove highly emotional comments from the dataset. This way we explore whether our classifiers perform better when trained on a less emotional dataset. The results of both approaches can be found in Chapter 4.

3 Dataset Exploration

The used dataset contains 212687 comments from the `r/AITA` subreddit with associated binary labels `YTA` and `NTA`. The comments typically consist of a final judgement abbreviated by keywords

like YTA, NTA, ESH, etc. as well as an explanation or reasoning behind the judgement. Because of this automatically self-labelling data, one could expect the labelling fairly precise relative to the author's intent. Solely the case of mistyping users decreases the accuracy of the labels.

When looking at the dataset, we found that the mapping of the larger set of possible user's judgemental keywords to the two binary labels is not trivial. Exploring this mapping function seems interesting for developing an understanding of the underlying data. We theorize that comments get mapped to the binary label NTA, if they contain it as a token. This theory proves to be correct: A scan of the entire dataset for YTA-labelled comments whose bodies contain the keyword NTA yields an empty set of outputted comments. However, there are a variety of samples which impede the formulation of some further simple mapping rules. The following example demonstrates this trait of the dataset:

> **Datapoint Example: (bold lettering for clarification only)**
> **Comment**: "Because of how you responded, **YTA**. You didn't need to be so judgemental about it. I get that you don't want to be consistently feeding them all but your reaction was rash and probably a little embarrassing. You didn't have to usher them out so fast."
> **Label**: NTA

Although this comment's semantics clearly indicate a typical user's perception of YTA and albeit the literal occurrence of YTA, the dataset labels this comment with NTA. This example shows that the underlying labelling process of the dataset might not be fully correct. Hence we conclude and acknowledge that the dataset's inaccuracies may have an impact on the validity our final results.

Another finding is, that the dataset is inbalanced as 150040 of the comments are labelled as NTA while 62647 are labelled as YTA. This inbalance can lead to biased results. Additionally it can be observed that the labels are already contained in the text of many comments. This can cause the model to train on the labels and not the textual data itself. Therefore, it is necessary to clean the data beforehand. This also extends to the other judgement phrases besides YTA and NTA (such as ESH/"Everyone Sucks Here" or NAH/"No Assholes Here"), although they do not count as labels of the underlying dataset themselves. This decision is based on the factor that even the other judgement phrases might distort the model due to their implicated relation towards the two main dataset labels.

Moreover, the cleaning preprocessing also consists of stripping special characters and white spaces from all tokens of all documents. Note that documents which consist entirely out of textual labels or special characters thus result in a document length of zero. Since an empty document retains no meaningful information, we decided to remove them from the dataset. Due to this, the initial scope of the dataset decreases by 5865 documents. The cleaned dataset's 206822 samples can be separated into 145971 NTA (4069 less than before) and 60851 YTA labelled datapoints (1796 less).

Figure 1 depicts the distribution of cleaned documents according to their length. Note that the graph has a logarithmic scaling to show the outliers (the documents with the highest token count). As one can see, the majority of documents contained in the cleaned dataset have less than 100 tokens

Supporting numerical data regarding the size of the cleaned documents can be inferred from table 1. max_tokens / min_tokens represents the largest/smallest document, as measured by the amount of tokens. Likewise, avg_tokens measures the average length across all documents and σ

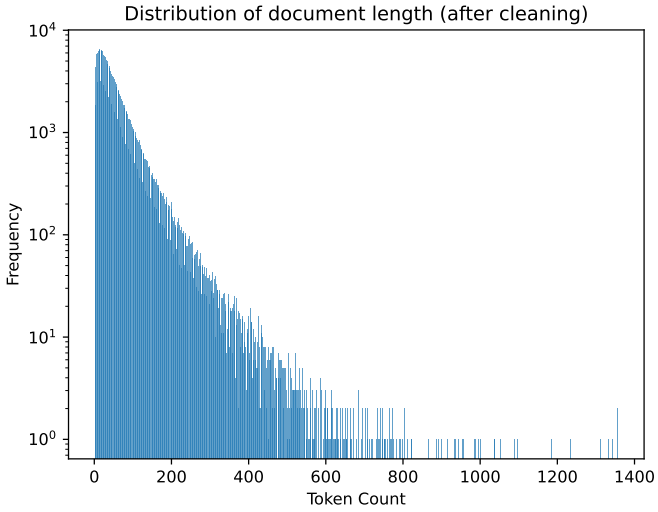


Figure 1: *Distribution of document length after cleaning.*

their standard deviation. Note that the removal of labels and special characters in the comments leads to a lower mean document length. The vocabulary size, abbreviated as `vocab_size` denotes the amount of unique tokens over all documents in the dataset. After removing stop words from the vocabulary, the resulting numbers are represented in the tabular field `vocab_size*`. Further explored aspects encompass the analysis of the most and least frequent words. It can be shown that the latter mostly appear be spelling errors, slang or neologisms. Furthermore, we found the correlation between document length and binary label to be rather low, being approximately 0.015.

Table 1: *Comparison between the original and cleaned dataset concerning the document's length.*

	min_tokens	max_tokens	avg_tokens	σ	vocab_size	vocab_size*
original data	1	1518	62.853	66.511	86137	85985
cleaned data	1	1357	55.557	58.823	77831	77679

4 Evaluation

In a fist step we evaluated the Chi-Squared test to check for a dependence between the `YTA/NTA` label and the sentiment labels. Regardless of using *VADER* or *Twitter-roBERTa-base for Sentiment Analysis* for the sentiment classification, a very low dependence between `YTA/NTA` labels and the sentiment labels were found: The Chi-Squared tests yielded a result of 0.00174 for *VADER* and 0.01339 for *Twitter-RoBERTa-base for Sentiment Analysis*, respectively. Additionally we analyzed the proportions of negative, neutral and positive annotations relative to the `YTA/NTA` Label. We

found out that for both sentiment classifiers their individual distribution of sentiment classes for the labels `YTA` and `NTA` is approximately the same.

Table 2: *Analyzing Sentiment Distribution: Proportions of Negative, Neutral, and Positive Sentiment Annotations relative to YTA/NTA Labels*

		negative	neutral	positive
RoBERTa	YTA	31.88%	66.23%	1.88%
	NTA	39.30%	59.61%	1.08%
VADER	YTA	11.35%	75.36%	13.29%
	NTA	11.94%	76.68%	11.38%

As a next step we evaluated combinations of models, embedding methods and data set filtering methods with regards to the accuracy and F1 score. The affiliated results are shown in Table 3. The table depicts the accuracy and F1 scores of the different model configurations (cf. chapter 2). Each classifier has been trained on (1) the unfiltered dataset as well as on the dataset after it has been filtered by the pretrained (2.a) RoBERTa or (2.b) VADER sentiment classifier. The threshold for filtering using VADER has been set to 0.21, while RoBERTA uses a threshold of 0.8. The training was employed on 80% of the data (the training set), whereas the evaluation metrics have been computed on the remaining 20% (test set). The computed evaluation metrics have been calculated using the test set. We executed the Logistic Regression pipelines for 30000 iterations. For the MLP, we chose a learning rate $\eta = 0.0025$, a 38 dimensional hidden layer and trained for 30 epochs. Due to limitations in time and hardware, the GRU and the BERT model have run for only one epoch each. The former works on a batch size of 1 and a hidden layer dimensionality of 5, whereas for the latter, we chose a batch size of 32, a learning rate $\eta = 5 \times 10^{-5}$. Note that the BERT model uses a maximum document length of 512 tokens, therefore documents with a greater length are truncated to 512 tokens. This should not influence the performance much as the majority of the data points are shorter than this anyways.

Table 3: *Performance across filtered and unfiltered training sets for our models (described in chapter 2), showing accuracy (Acc.) and F1-Score with either YTA as the true positive (F1-Y) or NTA (F1-N), all in percent. For the sentiment filtering we use the pretrained sentiment classifiers RoBERTa and VADER, respectively.*

		(1) Unfiltered			(2) Filtered					
					(2.a) RoBERTa			(2.b) VADER		
		Acc	F1-Y	F1-N	Acc	F1-Y	F1-N	Acc	F1-Y	F1-N
Log.Reg.	TF-IDF	76.97	50.09	85.03	76.96	44.74	85.54	78.37	53.00	85.95
Log.Reg.	GloVe	70.78	13.79	82.41	70.70	7.68	82.59	70.55	17.04	82.10
MLP	GloVe	70.70	36.35	80.97	72.25	33.27	82.48	71.17	37.33	81.28
GRU	GloVe	69.40	4.97	81.76	68.90	0.00	81.59	68.90	6.39	81.56
MLP	BERT	70.49	0.00	82.69	70.49	0.00	82.69	70.49	0.00	82.69

Contrary to our expectations, the baseline performed better than the more advanced models in both the unfiltered and the filtered cases. Rather than improving the accuracy and F1-scores,

GloVe and BERT embeddings decreased the scores in all cases. Generally speaking, while the F1 scores for the NTA class is rather high and comparable across the different model and embedding combinations, the F1 score for the YTA class is typically lower and varies a lot. The difference in values between F1-Y and F1-N could be explained by the NTA-to-YTA ratio, which is roughly 70% to 30%, though it is unclear as to why the YTA score is so volatile. Especially surprising is the performance of MLP in combination with BERT, as we hypothesized it to perform the best for sentences where order is very important. When considering the accuracy and both F1 scores, it is clear that the model is labelling every comment to the majority class NTA, though it is unclear as to why this is happening. This also explains why the the metrics remain the same when using the filtering methods. A similar phenomenon can also be observed when using GRU with Glove, where the F1-Y score is close or equal to zero.

Filtering the data set using RoBERTa mostly had a minor positive impact on the accuracy but also decreased the F1-Y scores by a considerable amount. VADER on the other hand proved to be more suited, increasing most of the values, albeit only slightly, and decreasing some by a negligible amount. Taking both filtering methods into account and ignoring some outliers, most of the metrics were only minimally changed, with the difference often being smaller than 1%. In analogy to our research goal, we conclude that filtering highly emotional comments does not have a significant impact on the training of our YTA/NTA classifiers.

In the end the best performing model is the logistic regression using TF-IDF as an embedding and the filtered dataset using VADER. Even though it beats the unfiltered baseline, the performance is still considerably worse compared to the accuracy, F1-Y and F1-N score of 86%, 80% and 90% from [EPT22] and 89.03%, 78% from [BGW22]. Though one should note that both of them used a different but similar data set.

5 Conclusion

In this work, we conducted research to examine the relation between YTA/NTA labelled comments within the r/AITA community, and their emotionality and sentiment. Throughout our investigation, we explored a range of models for our classifiers and embeddings, aiming to uncover insights into the correlation between emotional comments, sentiment analysis, and the labelling of YTA/NTA.

However, we encountered challenges that significantly impacted the results of our study. Hence, we acknowledge that the outcomes fall short of our initial expectations of beating our baseline. We suspect the following factors to contribute to the observed subpar performance of our classifiers: (1) inconsistencies in the labelling process of the underlying dataset (cf. chapter 3); (2) diverse comments written in non-standard language. The latter poses a challenge due to variations in the writing styles of the different users, their language usage, and potential spelling mistakes. These textual characteristics hinder the generalization ability of our models.

In light of these challenges, future work can address these limitations in order to improve the performance of our classifiers. Firstly, a refinement of the labelled dataset should be made to ensure accurate and consistent labeling to work with. Secondly, we expect better results by further fine-tuning the models. Additionally, we could query advanced language models, such as GPT 4.0, for the task. First experiments using GPT 3.5 showed promising results.

In conclusion, while our study encountered challenges and limitations, it contributes to the understanding of the dependency between `YTA/NTA` comments of the `r/AITA` community and sentiment analysis. By addressing the aforementioned factors and further exploring advanced models and techniques, future research can pave the way for more accurate and insightful classification of comments within online communities like `r/AITA`.

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